```
import numpy as np

# Given data
x = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([52, 55, 61, 70, 82])
```

```
from sklearn.linear_model import LinearRegression

model_A = LinearRegression()
model_A.fit(x, y)

beta_0 = model_A.intercept_
beta_1 = model_A.coef_[0]

print("Model A equation: y =", beta_0, "+", beta_1, "* x")
```

```
Model A equation: y = 41.50000000000001 + 7.499999999999998 * x
```

```
from sklearn.preprocessing import PolynomialFeatures

poly = PolynomialFeatures(degree=4)
x_poly = poly.fit_transform(x)

model_B = LinearRegression()
model_B.fit(x_poly, y)

coefficients = model_B.coef_
intercept = model_B.intercept_

print("Model B intercept:", intercept)
print("Model B coefficients:", coefficients)
```

```
Model B intercept: 51.99999999999997
Model B coefficients: [ 0.00000000e+00 -1.50000000e+00  1.50000000e+00  1.02140518e-14
 -6.66133815e-16]
```

```
x_test = np.array([[6]])
pred_A = model_A.predict(x_test)
print("Model A prediction for x=6:", pred_A[0])
```

```
Model A prediction for x=6: 86.5
```

```
x_test_poly = poly.transform(x_test)
pred_B = model_B.predict(x_test_poly)
print("Model B prediction for x=6:", pred_B[0])
```

```
Model B prediction for x=6: 96.99999999999994
```

```
from sklearn.metrics import mean_squared_error

y_pred_A = model_A.predict(x)
mse_A = mean_squared_error(y, y_pred_A)

print("Training MSE for Model A:", mse_A)
```

```
Training MSE for Model A: 6.299999999999992
```

```
y_pred_B = model_B.predict(x_poly)
mse_B = mean_squared_error(y, y_pred_B)

print("Training MSE for Model B:", mse_B)
```

```
Training MSE for Model B: 5.755529164492502e-28
```

Q - Which model has higher bias?

Answer: Model A (Linear Regression)

because Model A is too simple to capture the non-linear relationship between study hours and marks. Hence, it underfits the data and has high bias.

Q - Which model has higher variance?

Answer: Model B (Degree-4 Polynomial Regression)

because Model B is highly flexible and fits the training data very closely. With a small dataset, this makes it sensitive to noise, leading to high variance and overfitting.